Filmsamling / Klassediagram

Hel pil = felt · Stiplet pil = bruger · Hul trekant = implementerer | + offentlig − privat ~ pakkesynlig understreget = static

uden pakke
Main
~ main(args: String[]): void

ui
UserInterface
- controller: Controller
- scanner: Scanner
- dateFormatter: DateTimeFormatter

+ UserInterface()
+ startProgram(): void
- createMovie(): void
- printMovies(): void
- printMoviesByChoice(): void
- searchMovies(): void
- editMovie(): void
- deleteMovie(): void
- markAsWatched(): void
- findMoviesNotWatchedRecently(): void
- selectMovie(): Movie
- readInt(prompt: String): int
- readBoolean(prompt: String): boolean
- readDate(prompt: String): LocalDate
- readMovieYear(prompt: String): int
- saveMovies(): void

controller
Controller
- movieCollection: MovieCollection
- fileHandler: FileHandler
- changed: boolean

+ Controller()
+ addMovie(title: String, genre: String, director: String, yearCreated: int, lengthInMinutes: int, inColor: boolean, lastWatched: LocalDate): void
+ getMovies(): ArrayList<Movie>
+ searchMovies(searchtext: String): ArrayList<Movie>
+ editMovie(movie: Movie, title: String, genre: String, director: String, yearCreated: int, lengthInMinutes: int, inColor: boolean, lastWatched: LocalDate): boolean
+ deleteMovie(movie: Movie): boolean
+ markAsWatched(movie: Movie, date: LocalDate): void
+ getMoviesNotWatchedForDays(days: int): ArrayList<Movie>
+ loadMovies(): int throws FileNotFoundException
+ getSkippedLineCount(): int
+ saveMoviesIfChanged(): boolean throws FileNotFoundException
+ getMoviesSortedByTitle(): ArrayList<Movie>
+ getMoviesSortedBy(sortField: SortField): ArrayList<Movie>
- getComparator(sortField: SortField): Comparator<Movie>

domain
Movie
- title: String
- director: String
- yearCreated: int
- inColor: boolean
- lengthInMinutes: int
- genre: String
- lastWatched: LocalDate
- FIRST_MOVIE_YEAR: int {readOnly}

+ Movie(title: String, genre: String, director: String, yearCreated: int, lengthInMinutes: int, inColor: boolean, lastWatched: LocalDate)
+ hasBeenWatched(): boolean
+ markAsWatched(date: LocalDate): void
+ setLastWatched(lastWatched: LocalDate): void
+ getDaysSinceWatched(today: LocalDate): long
+ getTitle(): String
+ setTitle(title: String): void
+ getYearCreated(): int
+ setYearCreated(yearCreated: int): void
+ isInColor(): boolean
+ setInColor(inColor: boolean): void
+ getDirector(): String
+ setDirector(director: String): void
+ getLengthInMinutes(): int
+ setLengthInMinutes(lengthInMinutes: int): void
+ getGenre(): String
+ setGenre(genre: String): void
+ getLastWatched(): LocalDate
+ toString(): String
+ compareTo(other: Movie): int

1 movies 0..*

domain · «enumeration»
SortField
TITLE
DIRECTOR
YEAR
LENGTH
GENRE

domain
MovieCollection
- movies: ArrayList<Movie>

+ addMovie(title: String, genre: String, director: String, yearCreated: int, lengthInMinutes: int, inColor: boolean, lastWatched: LocalDate): void
+ addMovie(movie: Movie): void
+ getMovies(): ArrayList<Movie>
+ getNumberOfMovies(): int
+ searchMovies(searchText: String): ArrayList<Movie>
+ editMovie(movie: Movie, title: String, genre: String, director: String, yearCreated: int, lengthInMinutes: int, inColor: boolean, lastWatched: LocalDate): boolean
+ deleteMovie(movie: Movie): boolean
+ findMoviesNotWatchedRecently(limit: LocalDate): ArrayList<Movie>
+ getSortedMovies(comparator: Comparator<Movie>): ArrayList<Movie>
+ getMoviesSortedByTitle(): ArrayList<Movie>

java.lang · «interface»
Comparable<Movie>

filehandling
FileHandler
- HEADER: String {readOnly}
- fileName: String
- skippedLineCount: int

+ FileHandler(fileName: String)
+ saveMovies(movies: ArrayList<Movie>): void throws FileNotFoundException
+ loadMovies(): ArrayList<Movie> throws FileNotFoundException
+ getSkippedLineCount(): int
- toCsvLine(movie: Movie): String
- parseMovie(line: String): Movie

Sortering Controller vælger en comparator ud fra SortField. MovieCollection bruger den til at sortere filmene.

domain
TitleComparator
+ compare(movie1: Movie, movie2: Movie): int

domain
DirectorComparator
+ compare(movie1: Movie, movie2: Movie): int

domain
YearComparator
+ compare(movie1: Movie, movie2: Movie): int

domain
LengthComparator
+ compare(movie1: Movie, movie2: Movie): int

domain
GenreComparator
+ compare(movie1: Movie, movie2: Movie): int

java.util · «interface»
Comparator<Movie>